VMware to Huawei Cloud: AI-Assisted Migration



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AI-Assisted Migration: Usage and Disclaimer

Partner training | Huawei Cloud MaaS for VM to ECS migration

Sharing of capabilities, not a formal delivery commitment

Central Note

This session presents the AI-assisted capabilities of Huawei Cloud MaaS and their application ideas for migrating VMs from cloud to ECS. It does not process real customer data nor constitute a formal delivery plan, security or compliance commitment, or product feature promise.

1. Data and Privacy

Security / Compliance / Protection

- No real customer data is collected or processed
- Use simulated, test, or anonymized data
- Do not upload accounts, passwords, keys, certificates, or data

2. Usage Limits

Reference / Human Review

- AI supports planning, analysis, scripts, and risk assessment
- It does not replace design, expert review, testing, or decisions
- Review plans, scripts, and configurations before the project

3. Customer Authorization

Authorization / Approval

- The customer decides whether to use AI in real projects
- Without authorization, do not enter customer information into AI
- Respect security, compliance, confidentiality, and approvals

4. Not Recommended Scenarios

Prudent Evaluation / Manual Process

- Evaluate or avoid highly sensitive security scenarios
- Classified systems, financial core, defense, or energy
- Use manual migration under strict sovereignty or AI prohibition

AI-assisted migration improves efficiency; it does not replace human review or customer authorization. Real implementation must be based on customer confirmation, security, compliance, and project situation.

The Migration Challenge

Current Landscape

Migrating VMware VMs to Huawei Cloud ECS
requires careful planning and execution.

Typical scenario: 100 VMs
(60 Linux, 40 Windows), 20TB total disk.

Challenges: OS compatibility, network
config, downtime, data integrity.

AI Opportunity

AI-assisted analysis automates inventory,
dependency mapping, and risk assessment.

Reduces planning time from weeks to days.
Eliminates human errors in service mapping.

Generates optimal migration plans based
on workload priorities and dependencies.

Workshop Objective

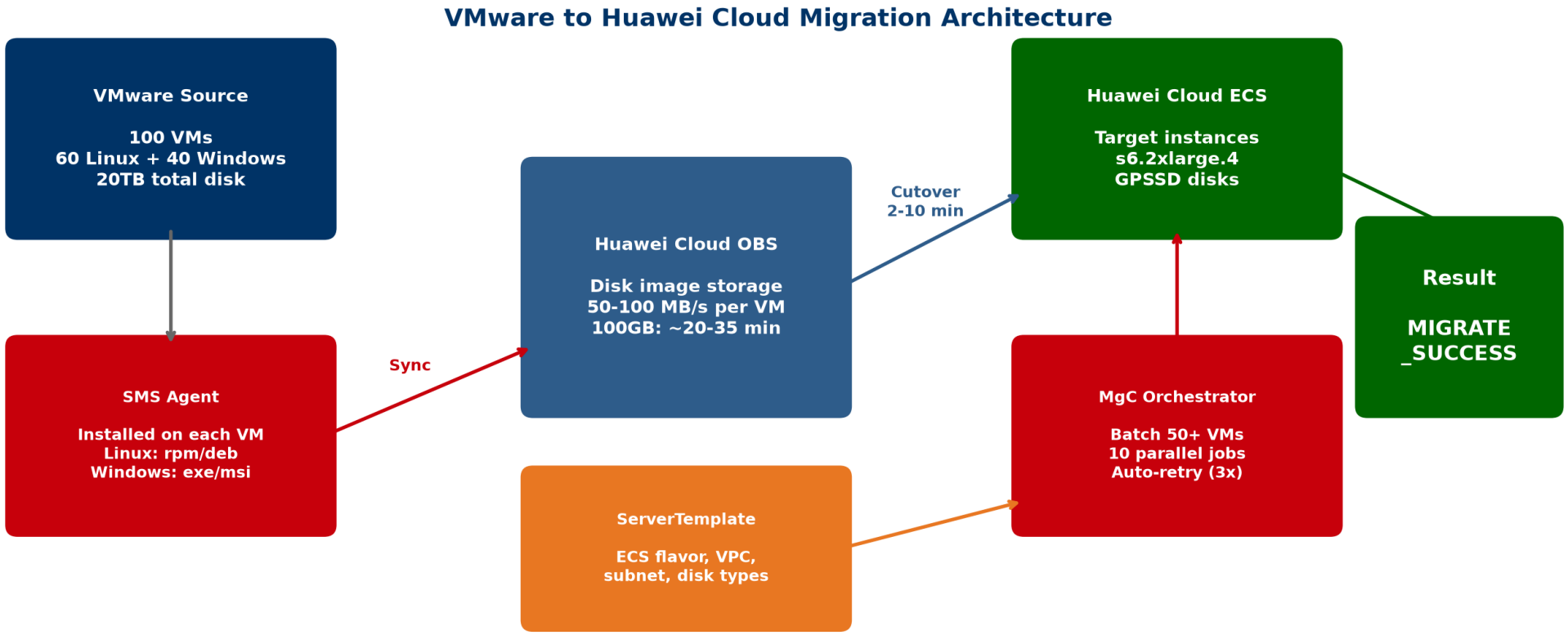
Demonstrate AI-assisted VMware to Huawei
Cloud migration with real field data.

Show SMS + MgC batch orchestration.
Present quantitative results from real cases.

Share 6-week production migration plan.

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Assessment & Discovery



Key metrics: 100 VMs (60 Linux, 40 Windows) | 20TB total disk | SMS agent: 2 weeks | MgC batch (10 parallel): 3 days

Steps: Inventory all source VMs -> Identify dependencies -> Categorize by priority -> Check OS compatibility -> Document network topology -> Estimate downtime

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SMS Agent & ServerTemplate

SMS Agent Installation

Download SMS Agent from Huawei Cloud console
Install on each source VM:
  Linux: rpm/deb package (5 min)
  Windows: exe/msi installer (10 min)
Configure with AK/SK + target region
Verify connectivity to SMS service (port 443)
Agent runs as daemon, auto-syncs disk to OBS
Monitor status: Active, Syncing, Error

Sync speed: 50-100 MB/s per VM
100GB disk: ~20-35 min sync time
Non-disruptive -- VM stays running

ServerTemplate Configuration

Define target ECS specs in SMS console
Select ECS flavor (match or exceed source):
  AWS m5.2xlarge (8vCPU/32GB)
  -> Huawei s6.2xlarge.4
Choose target VPC, subnet, security group
Select ECS image: public or private
Configure system disk: SSD/GPSSD/SATA
Set data disk mappings (source -> target)
Configure EIP, bandwidth, SG rules

Cost: $0.384/hr -> $0.234/hr = 39% savings
Disk: gp2 -> GPSSD (better performance)

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Migration Job Execution

Migration Job Execution: Two-Phase Process

Phase 1: Background Sync

- Agent uploads disk data to OBS
- Non-disruptive — VM stays running
- Sync speed: 50-100 MB/s per VM
- 100GB disk: ~20-35 min
- Monitor: sync %, data transferred, ETA

Ready →

Phase 2: Cutover

- Brief downtime: 2-10 min per VM
- Final incremental sync (<5% of disk)
- ECS creation: ~30 seconds
- Switch traffic to new ECS
- Decommission source after sign-off



On failure: review logs → fix issue → retry from last checkpoint

Huawei Proprietary - Restricted Distribution

Batch Migration with MgC

Orchestration

MgC for large-scale batch VM migrations
(50+ VMs).

Group VMs by application or priority tier.
Pre-migration checks: auto-validate OS,
disk, network compatibility.

Progress dashboard: aggregate view across
all VMs in migration group.

Parallel Execution

Configurable concurrency (e.g. 10 parallel).
50 VMs with 10 concurrency = 5 waves.
Total time: ~2 hours for 50 VMs.

Auto-retry on transient failures (3 attempts).
Pause/resume for maintenance windows.

First-pass success rate: >95%.

Auto-Validation

Post-migration auto-validation:
Boot check, network reachability, Service health.

Validation time: 30 min/VM manual, <5 min/VM automated via MgC.

Manual sign-off: 1 hour per app tier.
Update DNS, monitoring, CI/CD.

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Post-Migration Validation

Boot & Network

Verify ECS instance boot:
OS logs, systemd services, app startup

Network validation:
EIP reachable, internal connectivity, DNS resolution

Security: apply Huawei Cloud SGs, CTS audit, Anti-DDoS

Data & Performance

Data integrity:
Compare disk checksums or run application-level validation

Performance baseline:
CPU, memory, disk I/O vs source VM metrics

Validation: 30 min/VM manual, <5 min/VM automated

Cutover & Cleanup

Update DNS to point to new ECS
Update monitoring and CI/CD pipelines

Decommission source VM after sign-off
Retain backup per policy

Validation: ~30 min/VM manual, <5 min/VM automated via MgC
Manual sign-off: 1 hr per app tier

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Huawei Cloud Migration Tools

Service	Role	Migration Phase
SMS (Server Migration Service)	Agent-based disk sync & cutover	Core migration
MgC (Migration Center)	Batch orchestration & dashboard	Large-scale (50+ VMs)
ECS (Elastic Cloud Server)	Target compute instances	Post-migration
OBS (Object Storage)	Temporary disk image storage	During sync
VPC / Subnet / EIP	Network configuration	ServerTemplate setup
IMS (Image Management)	Custom OS images if needed	ServerTemplate setup
CES (Cloud Eye)	Monitoring & alerting	Post-migration
Terraform	Infrastructure as Code automation	All phases

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Real Case Results

Field Cases (2026)

Case 1 (Apr 18): la-north-2 -> la-south-2
  SMS migration: 16 min total
  Result: MIGRATE_SUCCESS

Case 2 (Apr 22): Mexico City2 -> Santiago
  SMS migration: MIGRATE_SUCCESS
  Postcheck: VPC, EIP, SG all passed

Case 3 (May 1): VMware -> la-north-2
  rsync fallback (SMS.6504)
  Full sync: 73 min (4391s)

Cost & Performance

Compute cost savings: 39%
  AWS m5.2xlarge: $0.384/hr
  Huawei s6.2xlarge.4: $0.234/hr

Cutover downtime: 2-10 min/VM
ECS creation: ~30 seconds
Sync speed: 50-100 MB/s per VM

Batch: 50 VMs, 10 parallel = ~2 hours
First-pass success rate: >95%

Error Handling (Proven)

SMS.6504: OS incompatible -> rsync
  Automatic fallback, no manual action

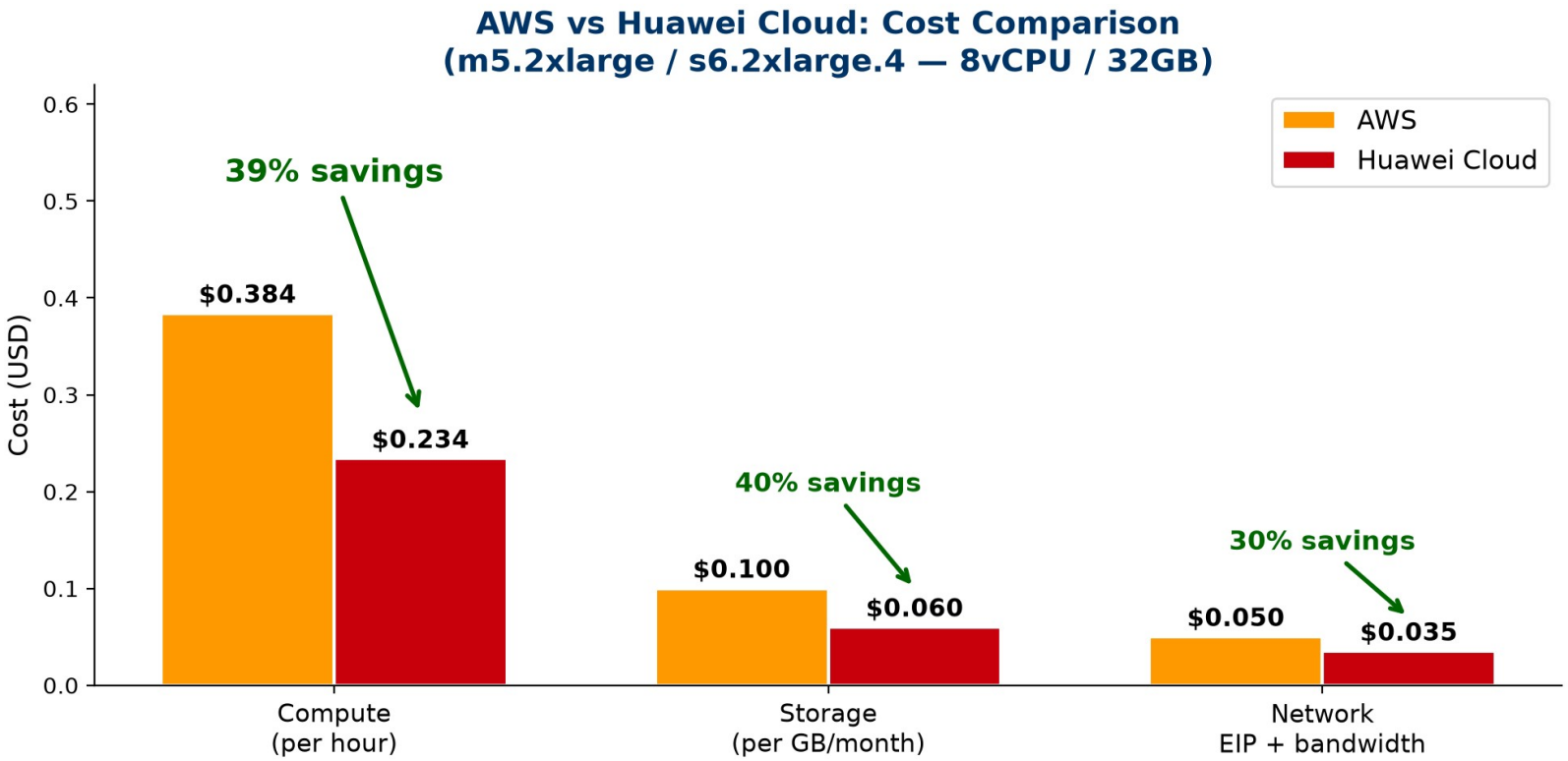
VPC.0114: Quota exceeded -> cleanup
  Delete unused VPC, retry

SMS.7605: Task residue -> cleanup
  Purge old tasks, retry

All errors have documented runbook fixes

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Cost Comparison



Flavor mapping: AWS m5.2xlarge (8vCPU/32GB) -> Huawei s6.2xlarge.4 | Disk: gp2 -> GPSSD | 39% compute savings | 40% storage savings

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AI-Assisted Benefits

Operational Efficiency

Reduction of time in analysis and planning
based on source infrastructure inventory.

Elimination of human errors in service
mapping and configuration.

Automation of repetitive migration tasks
across 50+ VMs in parallel.

Cost Optimization

39% compute cost savings demonstrated:
$0.384/hr (AWS m5.2xlarge)
-> $0.234/hr (Huawei s6.2xlarge.4).

Disk: gp2 -> GPSSD (better performance,
lower cost).

Reduced operational migration costs.

Risk Reduction

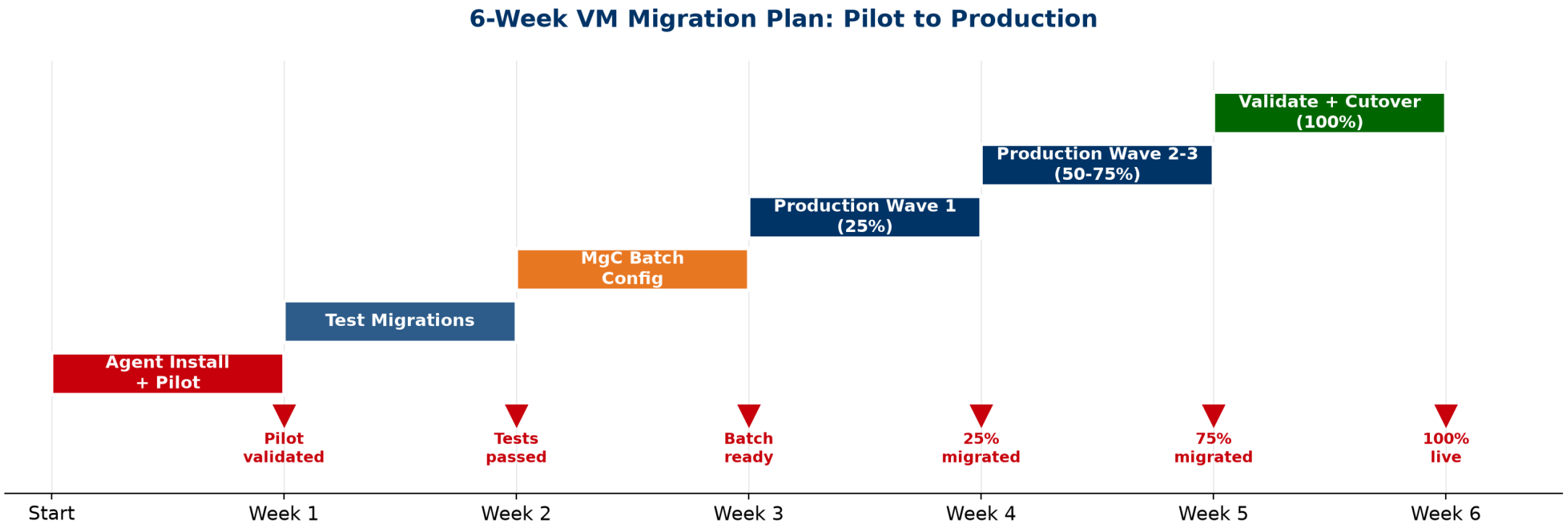
Early detection of risks, incompatibilities,
and critical dependencies.

Auto-fallback: SMS -> rsync when source
OS incompatible (SMS.6504).

Post-migration validation ensures data
integrity before production cutover.

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Next Steps: 6-Week Plan



Week 1: Agent install + pilot (2-3 non-critical VMs) | Week 2: Test migrations + validate end-to-end
Week 3: MgC batch config by app tier | Week 4-5: Production waves (25% per wave) | Week 6: Validate + cutover + decommission

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Resources & GitHub Links

VM Migration Skills
https://github.com/binrogithub/1-3-Cloud-Adoption-Skills/tree/main/AI-Assist-Migration/VM-Migration

SMS Migration
https://github.com/binrogithub/1-3-Cloud-Adoption-Skills/tree/main/AI-Assist-Migration/VM-Migration/huaweicloud-sms-migration

VMware Migration (MGC Cross-Region)
https://github.com/binrogithub/1-3-Cloud-Adoption-Skills/tree/main/AI-Assist-Migration/VM-Migration/Vmware-Migration

Skills include: SKILL.md, Terraform configs, Python scripts, runbooks, lessons-learned, migration bundles

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